

Central Tendency: Finding the Center of Data

Mathematics for AI/ML — Module 2: Probability & Statistics

Mean

The balance point

Median

The middle value

Mode

The most frequent

Data → Statistics → Machine Learning



Can ONE Number Represent 1,000 Values?

An ML model generates prediction errors for 1,000 customers. The raw output looks like this:

0.8, 1.2, 1.1, 0.9, 1.4, 2.3, 0.7, 1.6, 0.5, 1.9 ... (990 more values)

A data scientist needs to report performance to a stakeholder — not 1,000 numbers, just **one** meaningful summary. So here's the real question for today:

❏ "If I hand you 1,000 numbers, can you capture their essence in a single, meaningful value?"

Hold that thought. By the end of this lesson, you'll not only know the answer — you'll know *which* single number to pick, and *why* it matters for your models.

What Is Descriptive Statistics?

Every time you collect data — sensor readings, user clicks, exam scores, model errors — you're staring at a raw, unorganized pile of numbers.

Descriptive statistics gives you the tools to make sense of that pile.

"Descriptive statistics are methods used to summarize and describe the important characteristics of a dataset."

Three fundamental questions drive the field:

1. Where is the center?

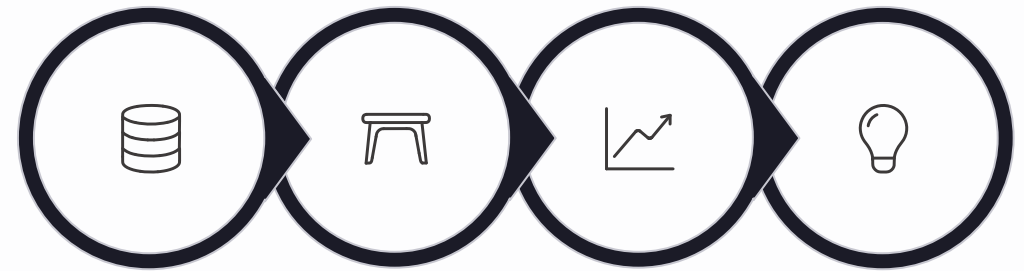
← TODAY's focus

2. How spread out is it?

Variance, Standard Deviation

3. What does it look like?

Shape, skewness, distribution



Raw Data

Organize

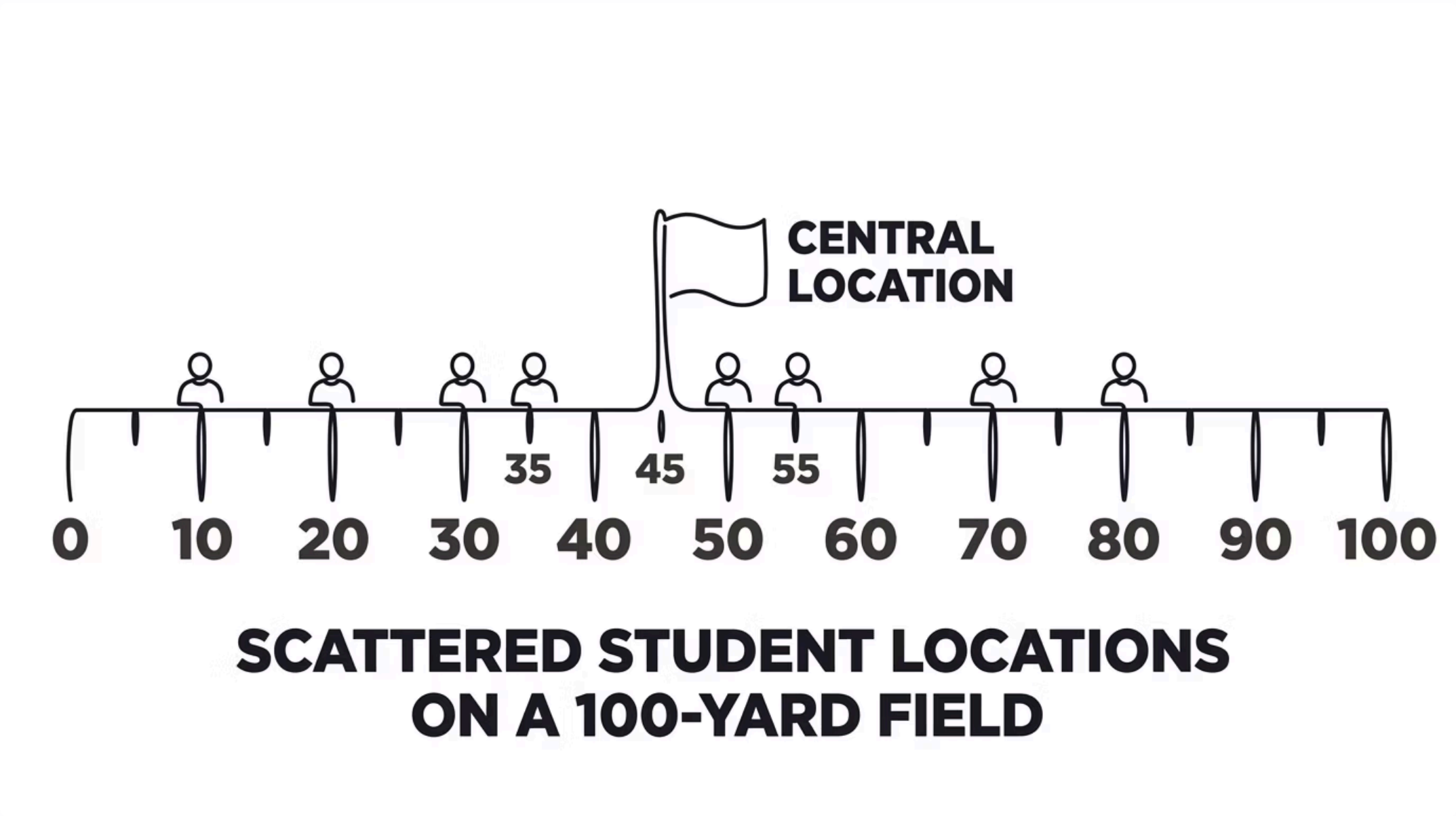
Summarize

Understand

What Does "Center" Even Mean?

Before we do any math, let's build the right intuition with a simple analogy.

Imagine students scattered across a football field at different yard lines. You need to plant **one flag** to represent where the group is located. Where do you put it?



That flag is **central tendency** — a single value that represents the typical or central location of a dataset. We have three different ways to find that flag:

Mean	Median	Mode
Mathematical balance	Middle position	Most common value

Three Ways to Find the Center

Same dataset. Three very different perspectives on where the "center" is. Let's see how they each look at the numbers **2, 3, 3, 4, 5, 7, 8**:



Mean = 4.57

The **mathematical balance point**. Add everything up and divide by count. If the data were weights on a ruler, this is where it would perfectly balance.



Median = 4

The **middle position**. Sort the values, find the one in the center. Half the data sits above it, half below — like the person standing exactly in the middle of an ordered queue.



Mode = 3

The **most frequent value**. Which value appears most often? In a bar chart, it's the tallest bar — the most popular choice in the dataset.

📌 These three give different answers for the same data. Choosing the right one is the skill we're building today.

Mean: The Balance Point

The Intuition

Place weights on a ruler, each representing a data value. The **mean** is the exact point where the ruler perfectly balances — neither side is heavier.

Sample Mean Formula

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Population Mean Formula

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

What each symbol means:

$\bar{x}$ (x-bar)

Sample mean — from a subset of the population

μ (mu)

Population mean — from the entire population

x_i

Each individual observation

n / N

Sample size / Population size

Σ (sigma)

Sum all values from $i = 1$ to n

i In ML, you almost always work with samples, so $\bar{x}$ is your everyday tool.

Mean: Step-by-Step Example

Dataset: 2, 4, 6, 8, 10

Step 1 – Sum all values:

$$2 + 4 + 6 + 8 + 10 = 30$$

Step 2 – Count observations:

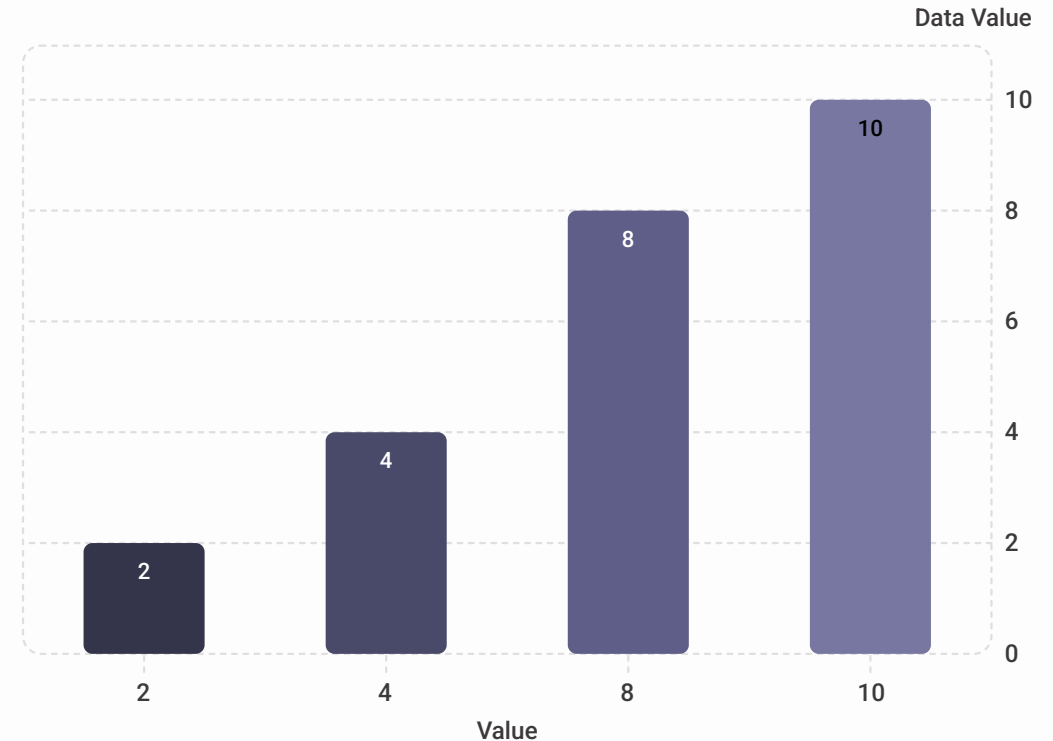
$$n = 5$$

Step 3 – Divide:

$$\bar{x} = \frac{30}{5} = 6$$

- ✔ Interpretation: The average value is 6. The dataset is perfectly balanced around this point – two values below (2, 4) and two above (8, 10).

Visualizing the Balance Point



Notice: the mean (6) sits exactly at the center – values spread **symmetrically** on both sides.

But There Is a Problem with the Mean...

Dataset A — No outlier:

10, 11, 12, 13, 14

$$\bar{x} = \frac{60}{5} = 12$$

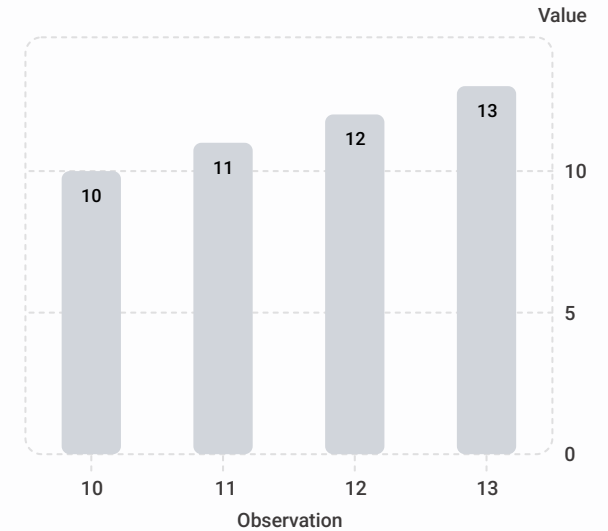
✓ Feels right. Typical value = 12.

Dataset B — One outlier:

10, 11, 12, 13, **100**

$$\bar{x} = \frac{146}{5} = 29.2$$

✗ Did the typical value suddenly become 29.2?



⚠ The mean is **sensitive to outliers**. A single extreme value pulls it far away from the true "typical" value — like a magnet dragging the balance point toward itself.

Median: The Middle Person

Imagine students lined up in order of height. The **median** is the student standing exactly in the middle — not the tallest, not the shortest, but the one with an equal number of students on each side.

Odd number of values:

Dataset: 1, 2, 3, **4**, 5, 6, 7

→ $n = 7 \rightarrow \text{middle position} = \frac{7+1}{2} = 4^{th}$

Median = 4 ✓

Even number of values:

Dataset: 1, 2, 3, **4, 5**, 6, 7, 8

→ $n = 8 \rightarrow \text{average of 4th \& 5th values}$

$$\text{Median} = \frac{4 + 5}{2} = 4.5$$

📌 **Golden Rule:** ALWAYS sort the data in ascending order FIRST — before locating the median. Forgetting to sort is the #1 student mistake.

Median: Step-by-Step Examples

Example 1 — Odd Count

Dataset: **9, 2, 7, 4, 5**

01

Sort ascending

2, 4, **5**, 7, 9

02

Find middle position

$n = 5 \rightarrow \text{position } \frac{5+1}{2} = 3$

03

Median = 5

2 values below · 2 values above 

Example 2 — Even Count

Dataset: **9, 2, 12, 7, 4, 5**

01

Sort ascending

2, 4, **5, 7**, 9, 12

02

Find two middle values

$n = 6 \rightarrow$ 3rd and 4th values: 5 and 7

03

Average them

$$\text{Median} = \frac{5 + 7}{2} = 6$$

Mean vs. Median: Who Handles Outliers Better?

Dataset: 10, 11, 12, 13, 100

Measure

Mean (pulled by outlier)

29

Median (stays near core)

12

Value

Mean = 29.2

Dragged far from the cluster of 10–13 by the outlier value of 100.
Misleading!

Median = 12

Sits right in the heart of the data. Unaffected by the extreme value of 100. ✓

📌 **Key Insight:** Mean is *sensitive* to outliers. Median is *robust* (resistant) to outliers. Neither is wrong — they answer different questions about the data.

Mode: The Most Frequent Value

The Intuition

A shoe store manager asks: *"Which shoe size should I order the most?"*
The answer isn't the average – it's the **most popular** size. That's the mode.

Dataset: 2, 3, 3, 3, 4, 5, 6

Mode = 3 (appears 3 times)

Unimodal

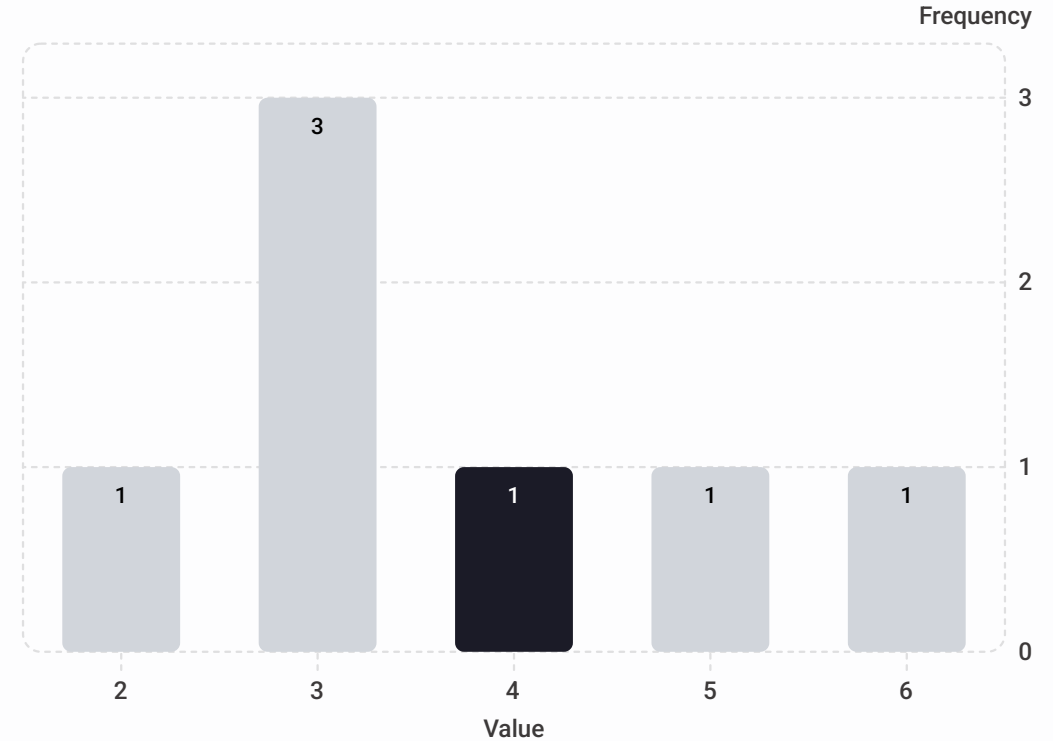
One mode: most common case

Bimodal

Two modes: e.g., {2, 2, 5, 5}

No Mode

All values appear equally: e.g., {1, 2, 3, 4}



The tallest bar = the Mode

i Mode is the **only** measure of central tendency that works with categorical data (e.g., colors, product types, city names).

The Three Measures Side by Side

Feature	Mean ($\bar{x}$ or μ)	Median	Mode
Meaning	Arithmetic average	Middle value (sorted)	Most frequent value
Calculation	$\Sigma x_i / n$	Sort → pick middle	Count frequencies
Sensitive to outliers?	✗ Yes — highly sensitive	✓ No — robust	✓ No — unaffected
Numerical data?	✓ Yes	✓ Yes	✓ Yes
Categorical data?	✗ No	✗ No	✓ Yes — only one that works!
Visual metaphor	 Balance scale	 Middle person in queue	 Tallest bar in chart

Which Measure Should I Use?

What does your data look like?

Symmetric, no strong outliers

Use MEAN

(e.g., Exam scores, temperature)

Skewed or has extreme outliers

Use MEDIAN

(e.g., Income, house prices)

Categorical or want most common value

Use MODE

(e.g., Most popular product, preferred color)

No single measure is universally best – context decides!

 Exam Scores

Likely symmetric → **Mean**

 House Prices

Skewed right → **Median**

 Incomes

Billionaires distort → **Median**

 Product Category

Categorical → **Mode**

Why Should an ML Engineer Care?

Before you train any model, you need to *understand* your data. Central tendency is your first diagnostic tool in Exploratory Data Analysis (EDA).



Model Error Evaluation

Mean Absolute Error (MAE) summarizes average prediction error. If one catastrophic error skews MAE upward, check the **median error** too — it reveals the typical case.



Feature Understanding

Customer income features often have billionaires as outliers. Using the **mean** blindly misrepresents typical customers. The **median** is more honest.



Categorical Features

For product preferences or user types, the **mode** tells you the dominant category — critical for default imputation and class-imbalance analysis.



Data Preprocessing

Missing value imputation strategies often use mean, median, or mode depending on the feature type and distribution — a core data-cleaning decision in every ML pipeline.

AI/ML Worked Example: Model Error Analysis

A regression model produces these **absolute errors** on a test set:

2, 3, 3, 4, 5, 3, 20

Step 1 — Calculate Mean:

$$\bar{x} = \frac{2 + 3 + 3 + 4 + 5 + 3 + 20}{7} = \frac{40}{7} \approx 5.71$$

Step 2 — Sort & Find Median:

Sorted: 2, 3, 3, **3**, 4, 5, 20

Median = 3 (4th value, $n = 7$)

What does the gap tell us?

Mean ≈ 5.71

Inflated by the single large error of 20. Looks like the model is performing worse than it really is.

Median = 3

The *typical* prediction is only off by 3 — much more representative of day-to-day model performance.

- ❏ **ML Takeaway:** A large gap between mean and median error signals the presence of outlier predictions worth investigating — perhaps data quality issues or edge cases.



Classroom Challenge

Dataset: 5, 6, 6, 7, 8, 8, 8, 9, 50

Work with the person next to you – **2 minutes** – and answer all five questions:

1

Calculate the Mean

Add all values, divide by count. What do you get?

2

Find the Median

Don't forget to sort first. Which is the middle value?

3

Find the Mode

Which value appears most frequently?

4

Which best represents "typical"?

Mean, Median, or Mode – and why?

5

What role does 50 play?

What effect does that value have on each measure?



Answers: Mean ≈ 11.9 · Median = 8 · Mode = 8 · Median/Mode best represent the data · 50 is an outlier that inflates the mean significantly.

Central Tendency — One-Page Summary

Formulas

Mean

$$\bar{x} = \frac{\sum x_i}{n} \quad \mu = \frac{\sum x_i}{N}$$

Median

Sort data. Pick middle value. If n is even, average the two middle values.

Mode

The value occurring most frequently. A dataset may have one, more than one, or no mode.

The Big Picture

→  Mean

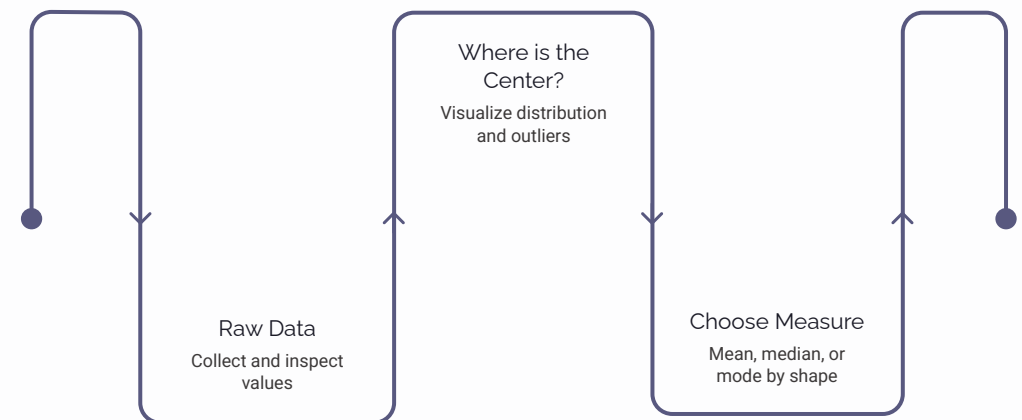
Balance point · Sensitive to outliers · Use for symmetric data

→  Median

Middle value · Robust to outliers · Use for skewed data

→  Mode

Most frequent · Works for categories · Use for nominal data



"Central tendency is not about blindly calculating an average. It is about finding a **meaningful** way to represent the center of data."